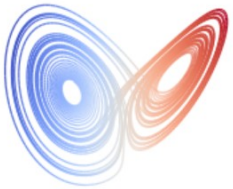
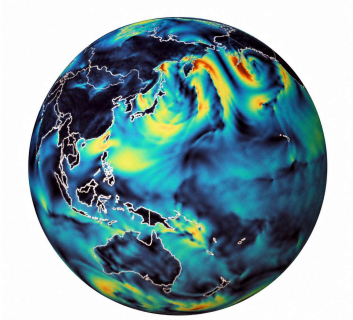
## Nonlinear Dynamical Systems & Geophysical Fields

Nonlinear dynamical systems

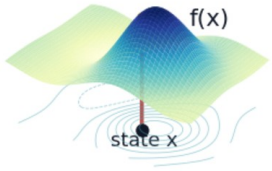


Lorenz 63 Attractor

Geophysical fields



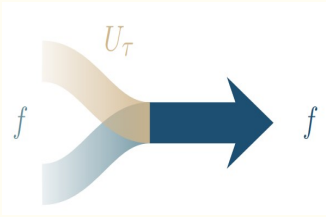
## Observables *f(x)*




$f(x): \mathbb{R}^d \to \mathbb{R}$

Observables are learned, not state trajectories.

## Learned Kolmogorov Backward Solution Operator $U\tau$

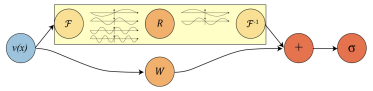



$U\tau\, f(x) = \mathbb{E}[f(X\tau)|X_0 = x]$

Evolves observables backward in time.

## Neural Operator Models

FNO (Fourier Neural Operator)



Neural Operator Models

GNO (Graph Neural Operator)



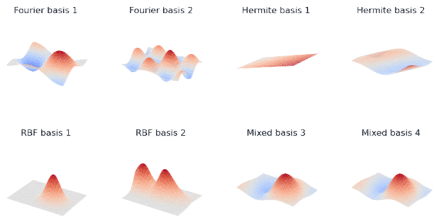
Others

Spherical FNO



## Koopman Dictionary & Koopman Matrix

Koopman Dictionary $\psi$

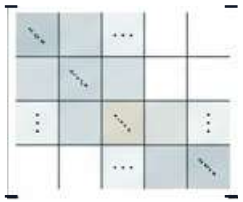



Galerkin Projection

$K_{ij} = \langle \psi_i, U\tau\, \psi_j \rangle$

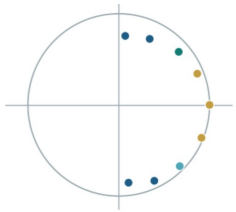
Koopman Matrix *K*

$U\tau\psi \approx K^T\psi$

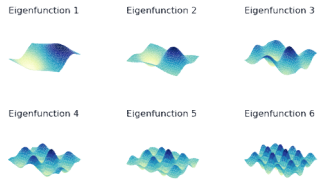


## Eigenvalues, Eigenfunctions & Stable Long-Term Forecasts

Koopman Eigenvalues



Koopman Eigenfunctions




Stable Long-Term Forecasts

